

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

Store A sells raspberries for **\$5.50** per pint and blackberries for **\$3.00** per pint. Store B sells raspberries for **\$6.50** per pint and blackberries for **\$8.00** per pint. A certain purchase of raspberries and blackberries would cost **\$37.00** at Store A or **\$66.00** at Store B. How many pints of blackberries are in this purchase?

Answer

- A. 4
- B. 5
- C. 8
- D. 12

Correct Answer: B

Rationale

Choice C is correct. It’s given that store A sells raspberries for **\$5.50** per pint and blackberries for **\$3.00** per pint, and a certain purchase of raspberries and blackberries at store A would cost **\$37.00**. It’s also given that store B sells raspberries for **\$6.50** per pint and blackberries for **\$8.00** per pint, and this purchase of raspberries and blackberries at store B would cost **\$66.00**. Let ***r*** represent the number of pints of raspberries and ***b*** represent the number of pints of blackberries in this purchase. The equation **$5.50r + 3.00b = 37.00$** represents this purchase of raspberries and blackberries from store A and the equation **$6.50r + 8.00b = 66.00$** represents this purchase of raspberries and blackberries from store B. Solving the system of equations by elimination gives the value of ***r*** and the value of ***b*** that make the system of equations true. Multiplying both sides of the equation for store A by **6.5** yields **$(5.50r)(6.5) + (3.00b)(6.5) = (37.00)(6.5)$** , or **$35.75r + 19.5b = 240.5$** . Multiplying both sides of the equation for store B by **5.5** yields **$(6.50r)(5.5) + (8.00b)(5.5) = (66.00)(5.5)$** , or **$35.75r + 44b = 363$** . Subtracting both sides of the equation for store A, **$35.75r + 19.5b = 240.5$** , from the corresponding sides of the equation for store B, **$35.75r + 44b = 363$** , yields **$(35.75r - 35.75r) + (44b - 19.5b) = (363 - 240.5)$** , or **$24.5b = 122.5$** . Dividing both sides of this equation by **24.5** yields **$b = 5$** . Thus, **5** pints of blackberries are in this purchase.

Choices A and B are incorrect and may result from conceptual or calculation errors. Choice D is incorrect. This is the number of pints of raspberries, not blackberries, in the purchase.